

AI Literacy

The competence to use, interpret, evaluate and question AI systems in a working context

GLOSSARY 02 - SHEET 1 / 8

PROFESSIONAL DEFINITION

AI literacy is the set of competencies that enables a person to use artificial intelligence systems knowledgeably, to interpret and evaluate their outputs, to communicate about them accurately, and to judge when a system should not be relied upon. Since 2 February 2025 it carries a legal meaning in the European Union: Article 4 of the AI Act requires providers and deployers to ensure a sufficient level of AI literacy among staff operating AI systems on their behalf (Long and Magerko, 2020; Regulation (EU) 2024/1689, Article 4).

WHO DEFINED IT ORIGINALLY

The operational competency definition was proposed by Duri Long and Brian Magerko (2020) at the Georgia Institute of Technology, who synthesised seventeen competencies and fifteen design considerations from a review of the computational- and data-literacy literatures. Ng, Leung, Chu and Qiao (2021) extended it into an educational taxonomy. The legal obligation derives from Article 4 of Regulation (EU) 2024/1689.

WHY THIS DEFINITION WAS DEVELOPED

Organisations were responding to AI by training either everyone in prompting or a few specialists in modelling, and neither addressed the actual failure mode: managers who could not evaluate an output they had been given. A competency-based construct made it possible to specify what a non-technical employee must be able to do, and the EU subsequently made that specification a compliance duty rather than a development preference.

FIGURE 1 - FOUR LEVELS OF AI LITERACY

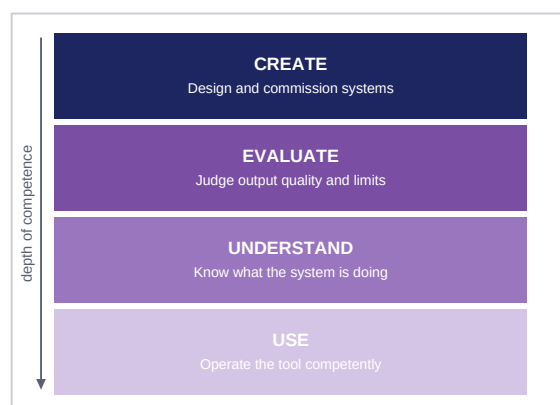


Figure 1. Literacy is cumulative. Most organisational programmes deliver the bottom layer and report the second; Article 4 of the AI Act is satisfied only where staff can perform the evaluation layer for the systems they operate (after Long and Magerko, 2020).

HOW IT IS USED

Map required competencies by role rather than by seniority, assess the current level against them, and close the gap with role-specific curricula; document the programme, since Article 4 compliance must be evidenced.

WHEN IT IS USED

Before granting staff access to an AI system, at onboarding, when a new class of system enters the organisation, and at the annual compliance review required under the AI Act.

BY WHOM IT IS USED

Human resources and learning functions designing the curriculum, line managers certifying role readiness, compliance officers evidencing Article 4, and every employee whose work involves an AI output.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor construct was digital literacy, which concerned the operation of tools whose behaviour was deterministic and inspectable. AI literacy adds the competencies that probabilistic and opaque systems require - recognising the limits of training data, interpreting confidence, identifying plausible-sounding error - and it locates the critical competence in evaluation rather than operation. Article 4 converted it from a capability model into an enforceable obligation, which no prior literacy construct had.

RELATED DEFINITIONS IN THIS COURSE

Artificial Intelligence - Unit 1

Human-in-the-Loop - Unit 2

Algorithm Aversion - Unit 2

Reskilling and Upskilling - Unit 4

EU AI Act - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

European Parliament and Council (2024) Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence, Article 4. Official Journal of the European Union, L, 12 July 2024.

Long, D. and Magerko, B. (2020) 'What is AI literacy? Competencies and design considerations', in Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York: ACM, pp. 1-16.

Ng, D.T.K., Leung, J.K.L., Chu, S.K.W. and Qiao, M.S. (2021) 'Conceptualizing AI literacy: an exploratory review', Computers and Education: Artificial Intelligence, 2, 100041.